

Logical Agents

Outline

I. Knowledge-based agents

II. The Wumpus world

III. Logic

I. Knowledge-Based Agents

- ♣ Problem solving agents do not know general facts.

An 8-puzzle agent does not know that two tiles cannot occupy the same space.

- ♣ Their atomic representations are very limited.

e.g., a list of all possible concrete states.

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- ♦ Intelligent agents need *knowledge about the world* in order to carry out reasoning for good decision making.

- Represent states, actions, etc.
- Incorporate new percepts.
- Update internal representation of the world.
- Deduce hidden properties of the world.
- Deduce appropriate actions.

Knowledge Base

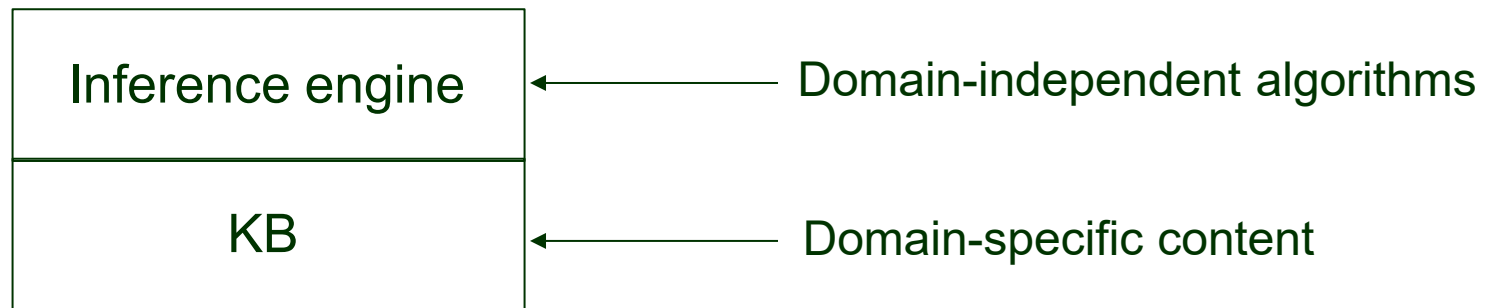
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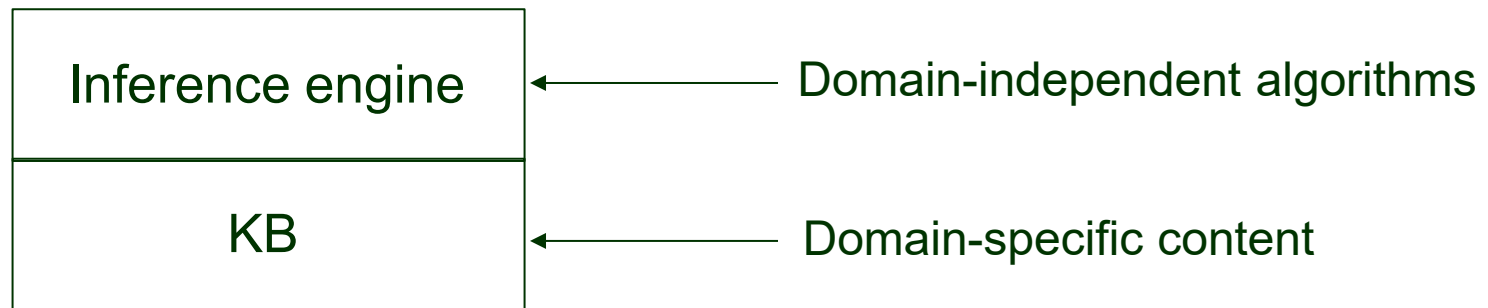
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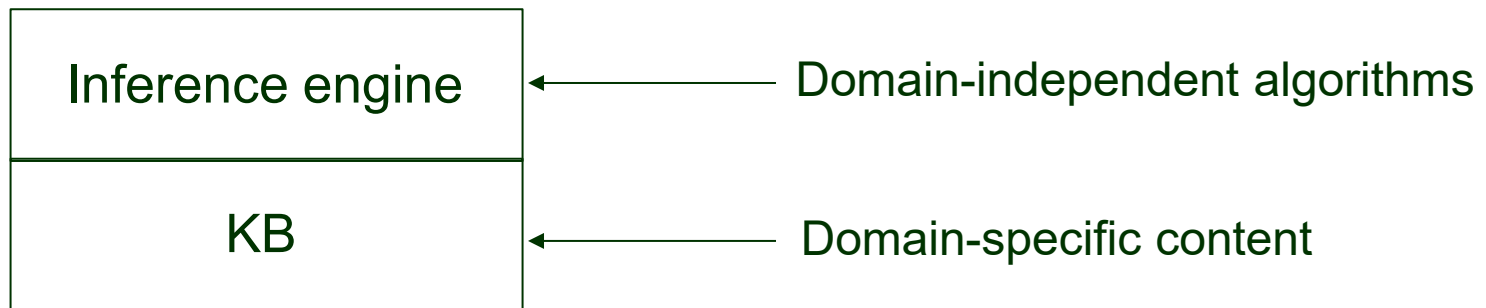


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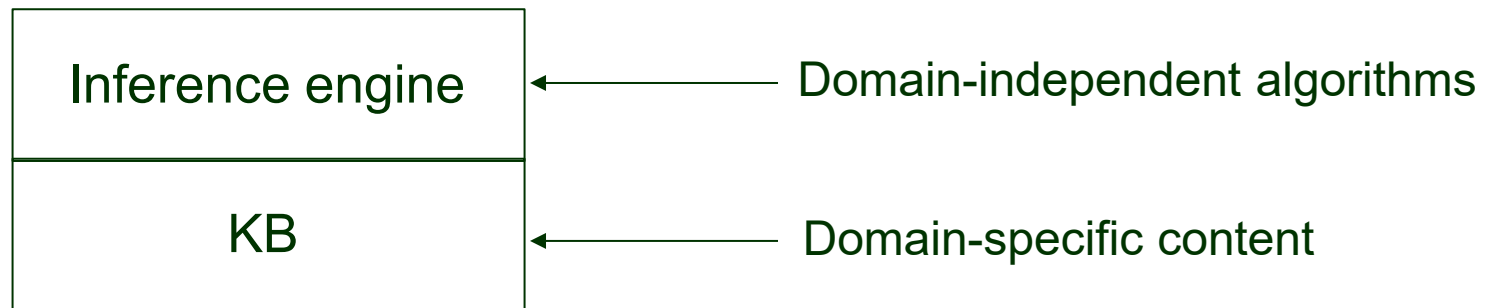
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TELL: Add new sentences to the KB.

Ask: Query the KB.

Inference: Derive new sentences from old.

Generic Knowledge-Based Agent

function KB-AGENT(*percept*) **returns** an *action*

persistent: *KB*, a knowledge base

t, a counter, initially 0, indicating time

TELL(*KB*, MAKE-PERCEPT-SENTENCE(*percept*, *t*))

action $\leftarrow$ ASK(*KB*, MAKE-ACTION-QUERY(*t*)) // asks what action
// it should perform.

TELL(*KB*, MAKE-ACTION-SENTENCE(*action*, *t*)) // tells what action
// was chosen.

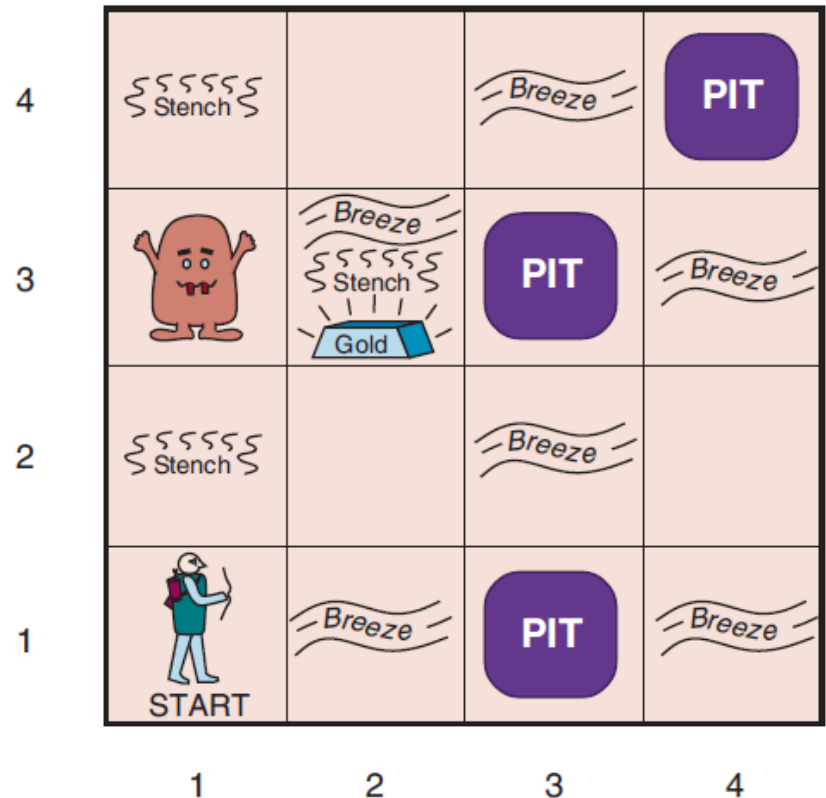
t $\leftarrow$ *t* + 1

return *action*

II. The Wumpus World

Cave consisting of connected rooms.

- Some rooms contain pits that will trap whoever enters them.
- The wumpus lurks in one room ready to eat whoever enters the room.
- The wumpus can be shot by the agent, who has **only one** arrow.
- A heap of gold is in a different room than where the wumpus lurks.

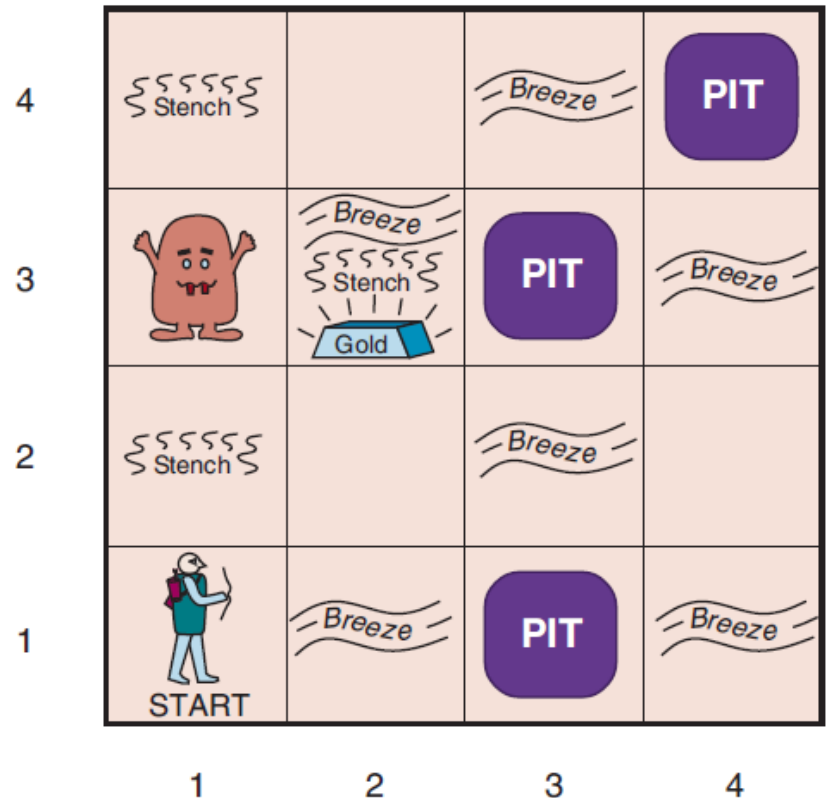


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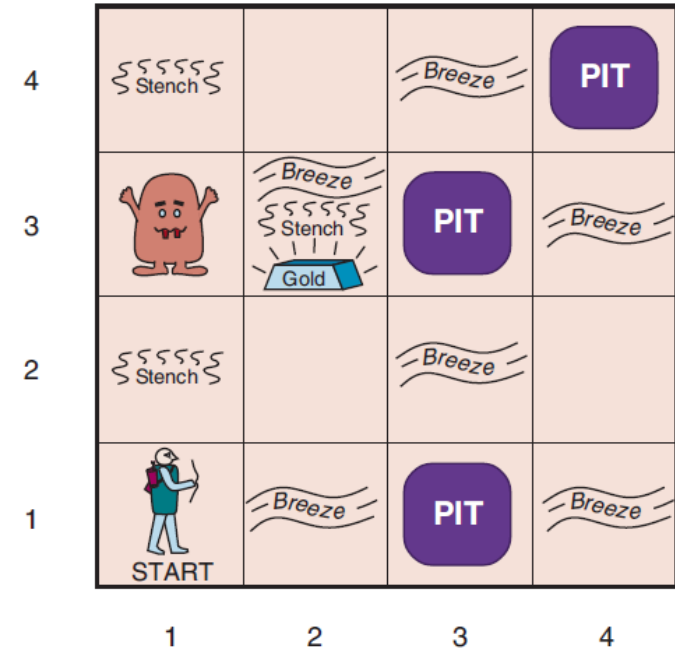
Cave

Goal: Find the gold and bring it back to the start without getting killed.

Task Environment

Performance measure

- +1000 (climbing out of the cave with the gold)
- -1000 (falling into a pit or being eaten by the wumpus)
- -1 (each action taken)
- -10 (using up the arrow)



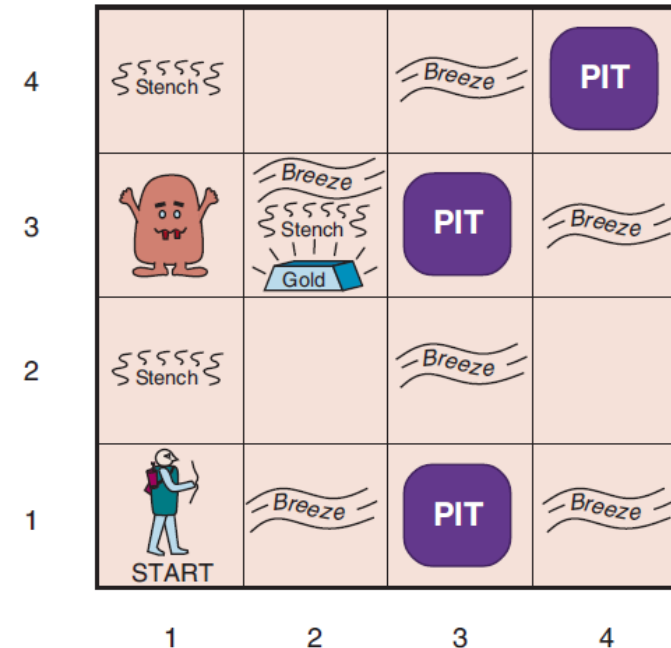
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Environment

- 4×4 grid surrounded by walls
- [1,1]: the starting square for the agent, who faces east
- locations of the gold and the wumpus:
 - $\neq [1, 1]$
 - otherwise randomly generated under uniform distribution (two distinct locations)
- 0.2 probability for a square other than [1, 1] and without gold or wumpus to be a pit



Actuators

Actuators:

1) *Forward*, *TurnLeft* by 90°, *TurnRight* by 90°

- Death of the agent if it enters a square containing a pit or a live wumpus.
- No movement if bumping into a wall.

2) *Grab*

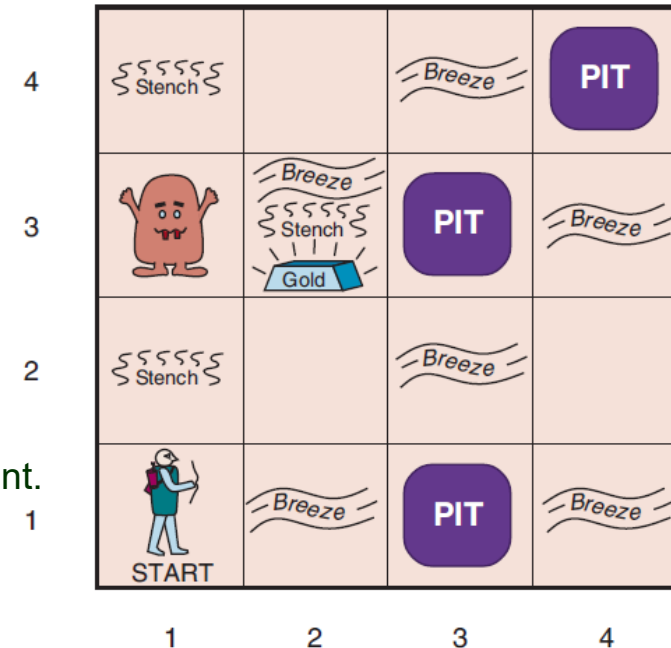
- Picks up the gold if it is in the same square as the agent.

3) *Shoot*

- Fire an arrow in the direction the agent is facing.
- The arrow continues until hitting the wumpus (who gets killed consequently) or a wall.

4) *Climb*

- Climb out of the cave if at [1, 1].



Sensors

5 sensors, each providing one bit of information:

1) *Stench*

- in the squares directly (not diagonally) adjacent to the wumpus

2) *Breeze*

- in the squares directly (not diagonally) adjacent to a pit

3) *Glitter*

- in the square where the gold is

4) *Bump*

- when the agent walks into a wall

5) *Scream*

- when the wumpus is killed

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Percepts in the form of a 5-vector:

e.g., [*Stench*, *Breeze*, *None*, *None*, *None*]

Characteristics of WW

- ◆ Deterministic, discrete, static, and single-agent

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Outcome specified.

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Locations of the pits and the wumpus are unknown.

Challenge: The agent needs to use logical reasoning to overcome its initial lack of knowledge about the environment's configuration.

Solution by a Knowledge-Based Agent

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
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[Stench, Breeze, Glitter, Bump, Scream]

Percept: [None, None, None, None, None]

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[1,2] and [2, 1] are free of dangers.

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Forward



1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

[Stench, Breeze, Glitter, Bump, Scream]

Percept: [None, None, None, None, None]



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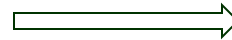
A pit in [1,1] , [2,2] , or [3, 1].

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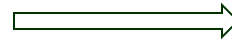
A pit in [2,2] or [3, 1].

Solution by a Knowledge-Based Agent

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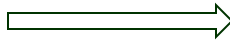
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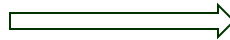
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[1,1] is OK

[2,2] is impossible because no stench was detected at [2, 1].

Only one unexplored square [1,2] is OK.

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Next Step

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More Inference

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
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[Stench, None, None, None, None]



The wumpus is in [1, 3].

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[Stench, None, None, None, None]



The wumpus is in [1, 3].



No breeze in [1,2].

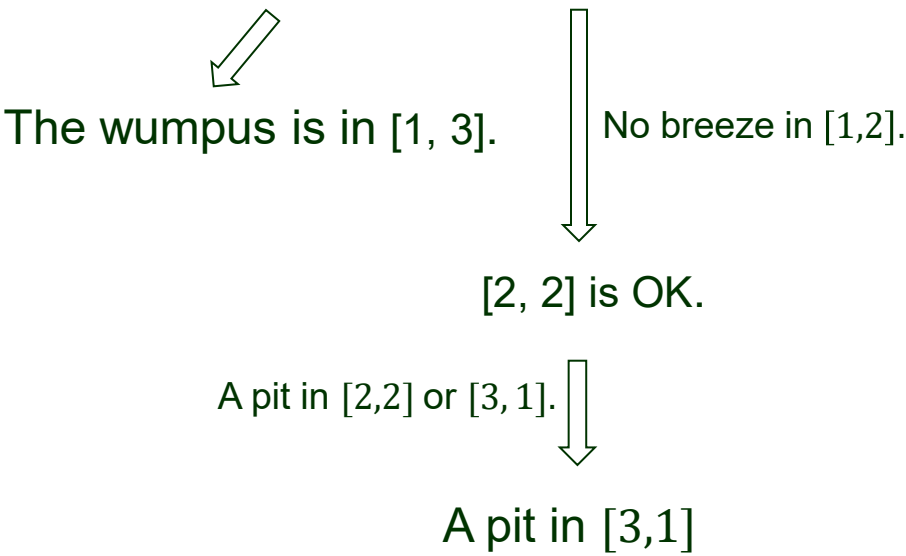
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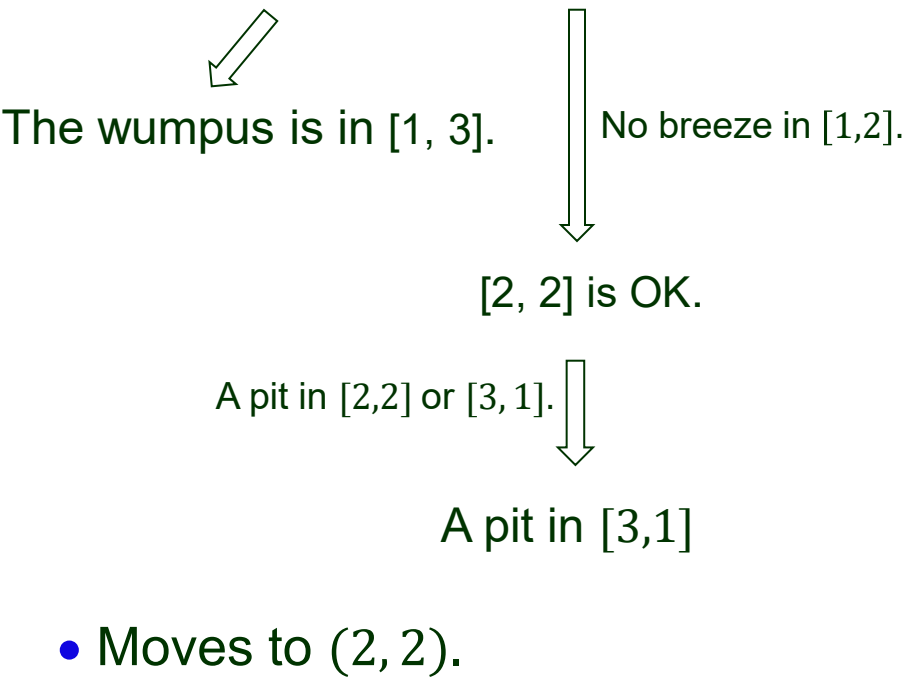


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1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

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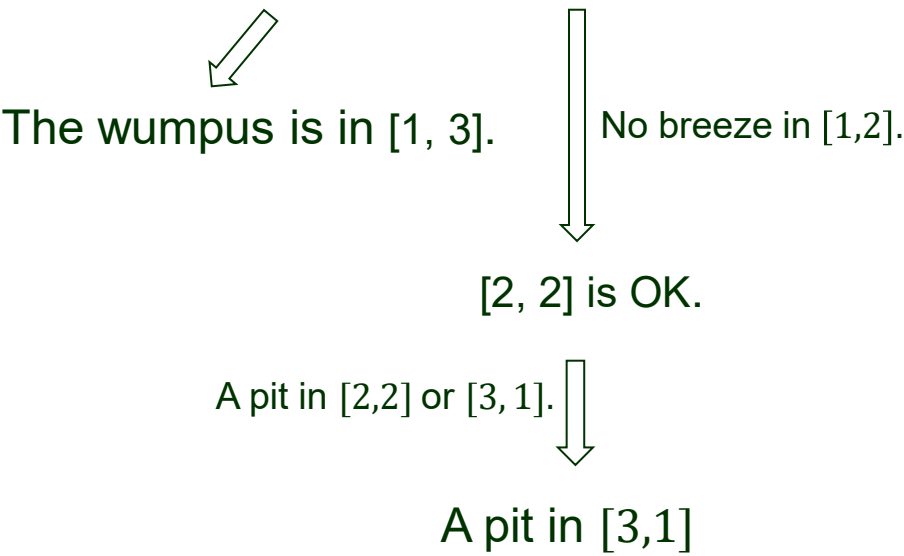


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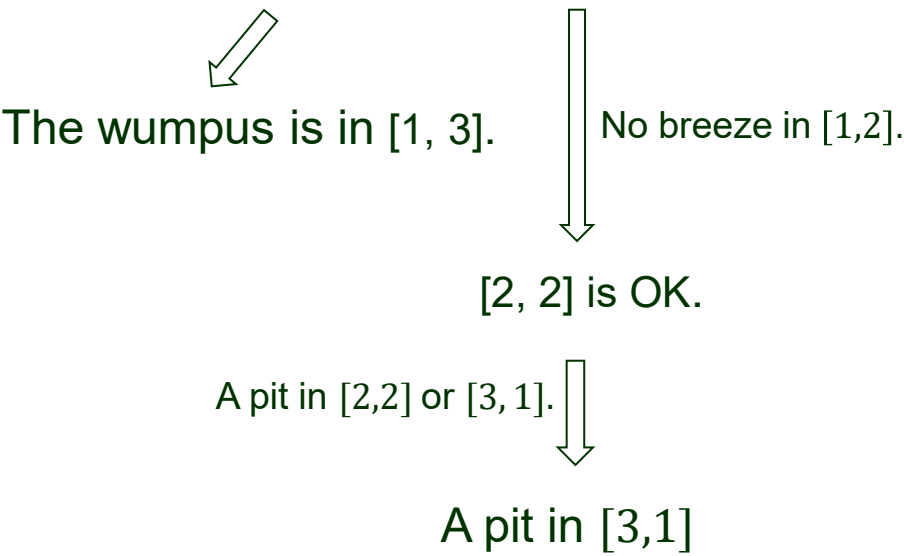
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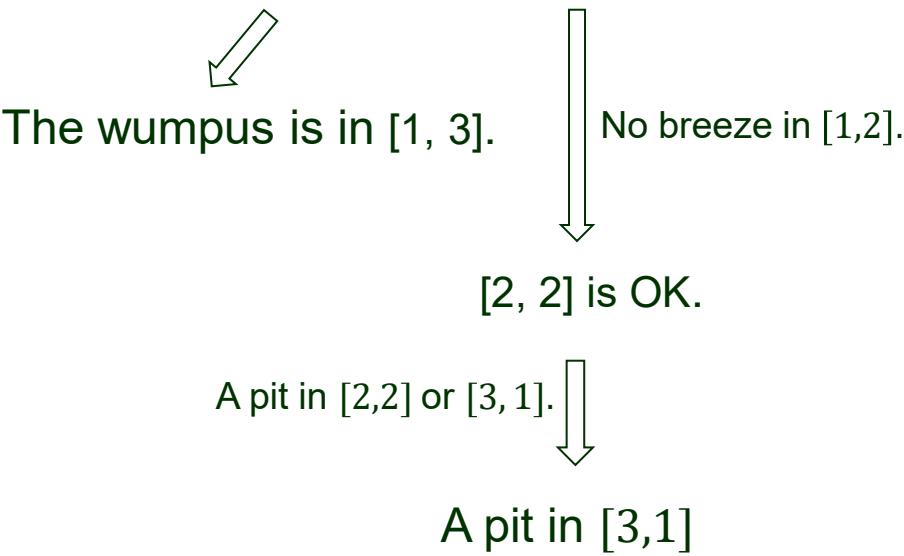
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A conclusion drawn is guaranteed if the available information is correct.

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Every sentence must be either true or false in each possible world.

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Example $x = 0$ entails $xy = 0$.

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1,3	2,3	3,3	4,3
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Knowledge base (KB) includes

- All the rules.
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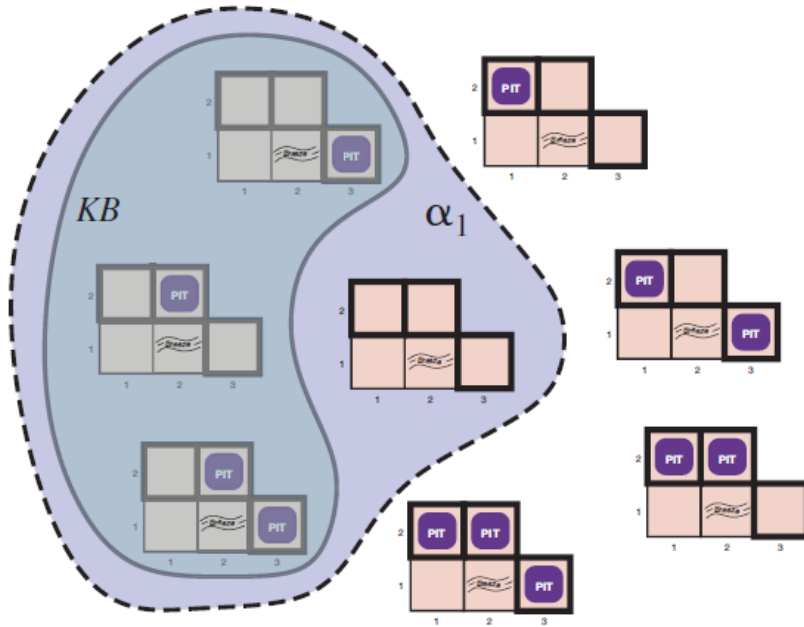
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8 possibilities if ignoring the KB.

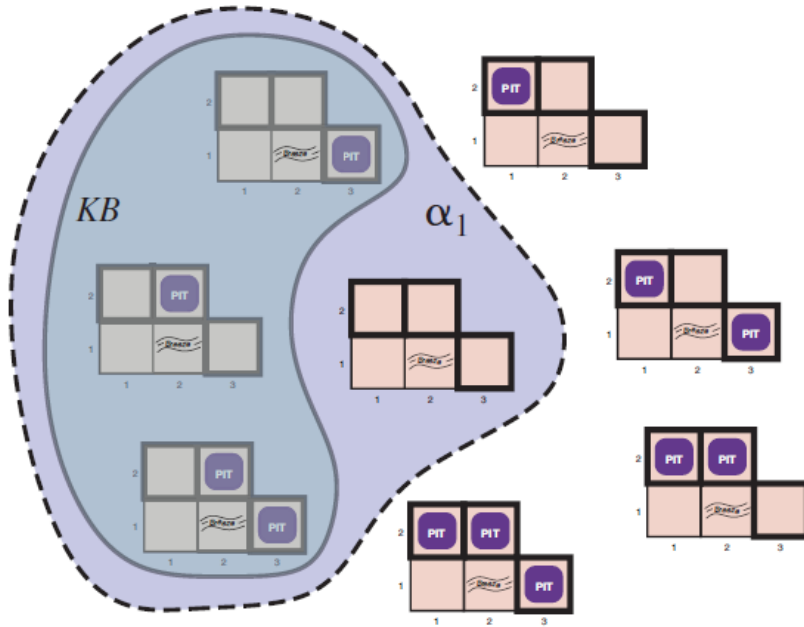
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8 possible models for the presence of pits in squares [1,2], [2,2], and [3,1].

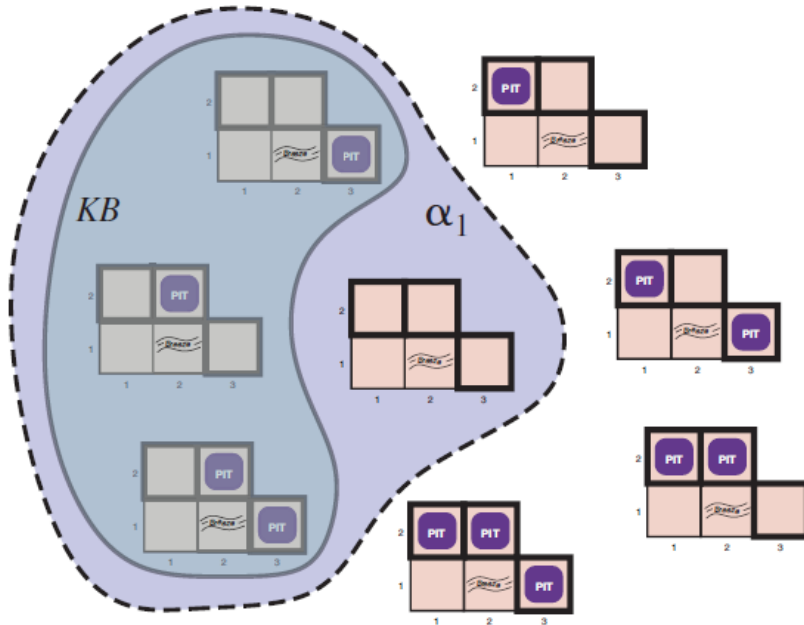
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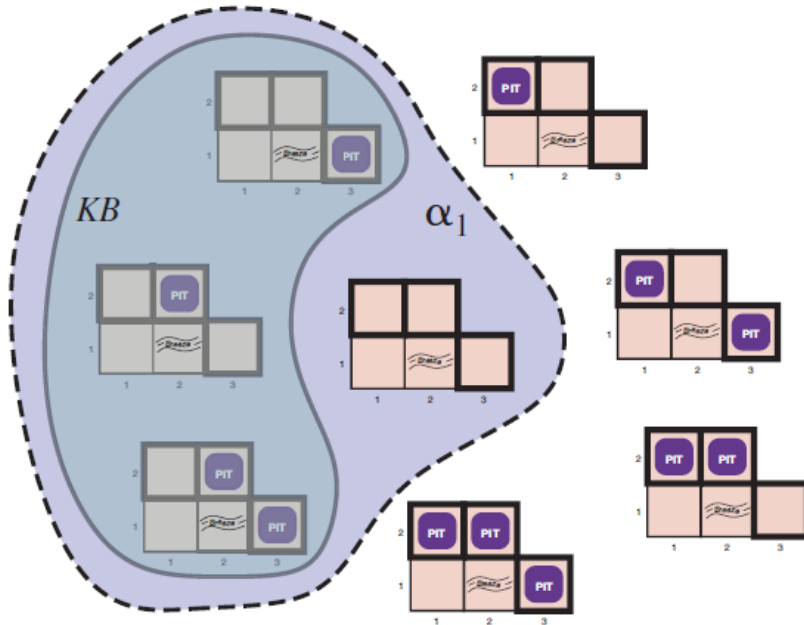


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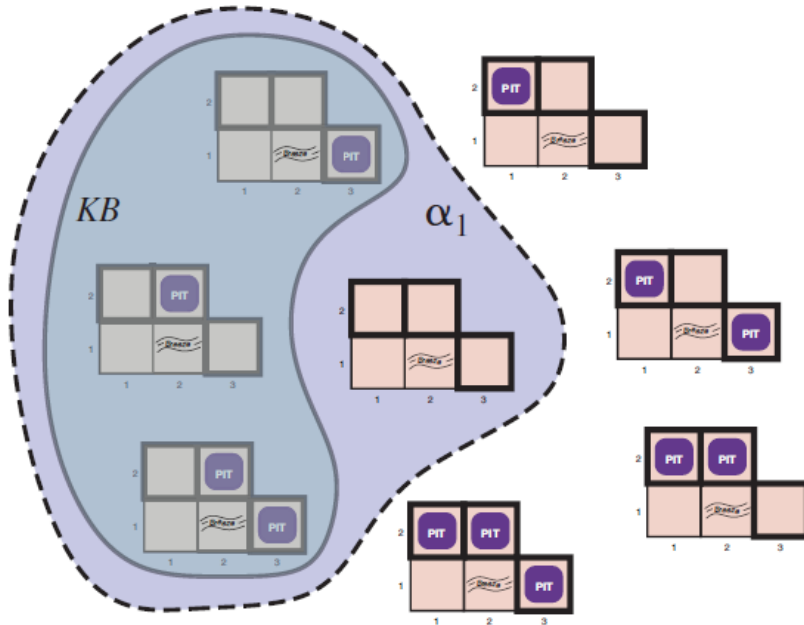
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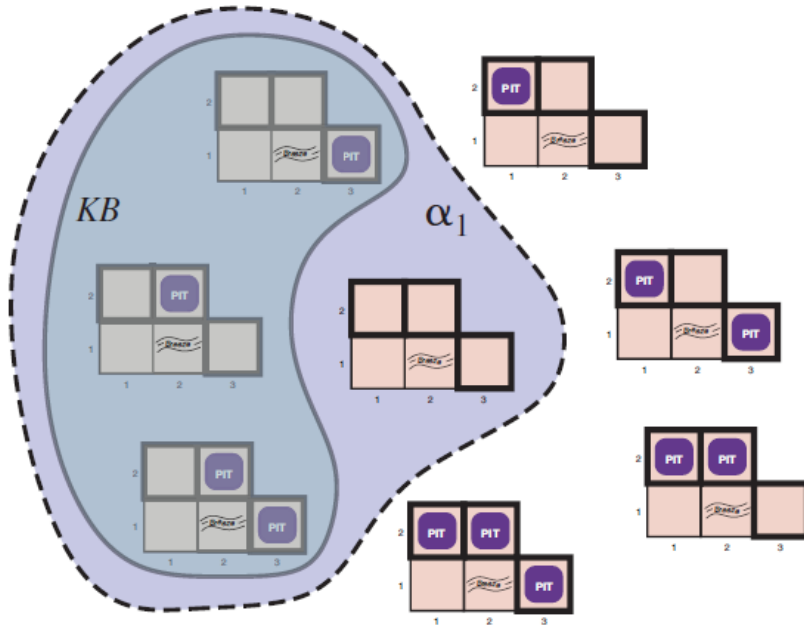
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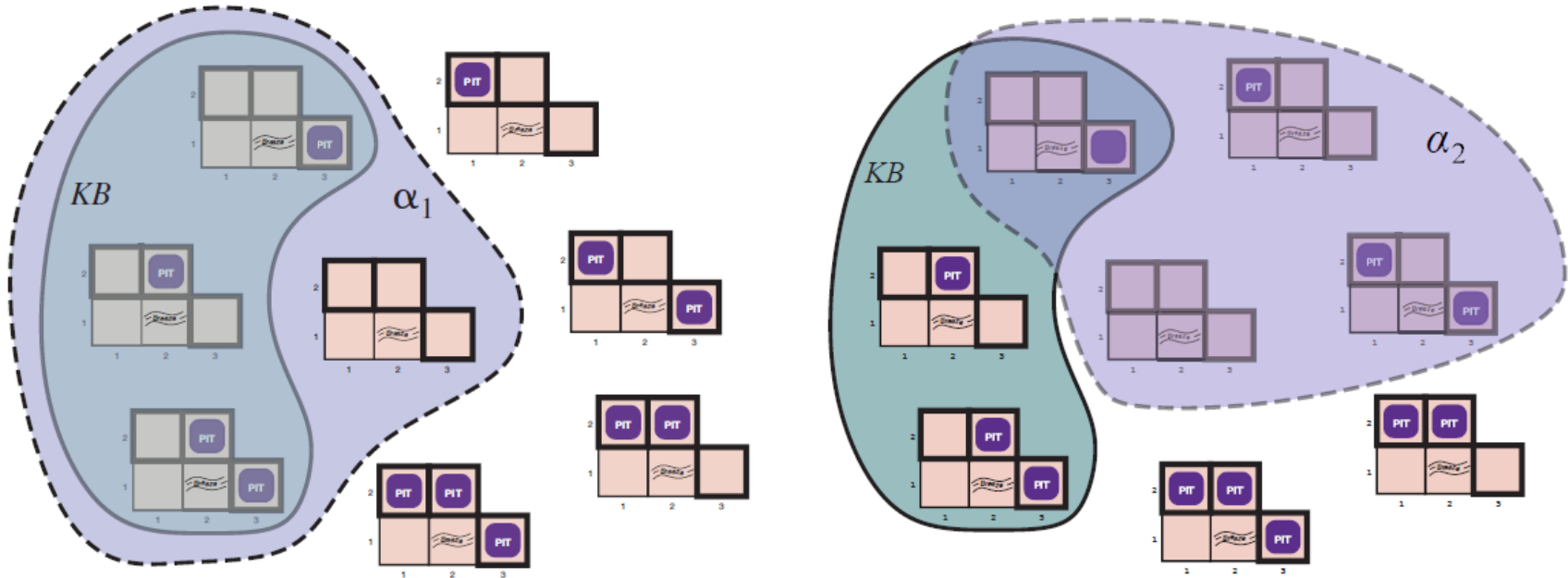
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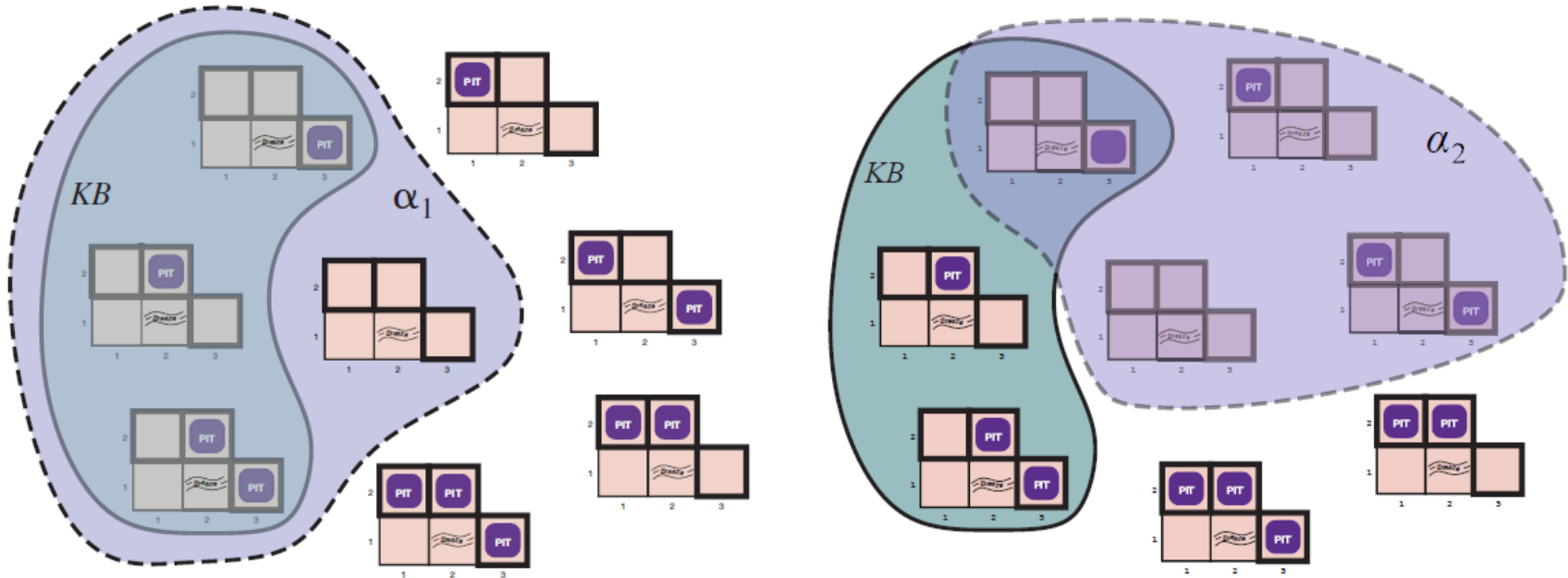
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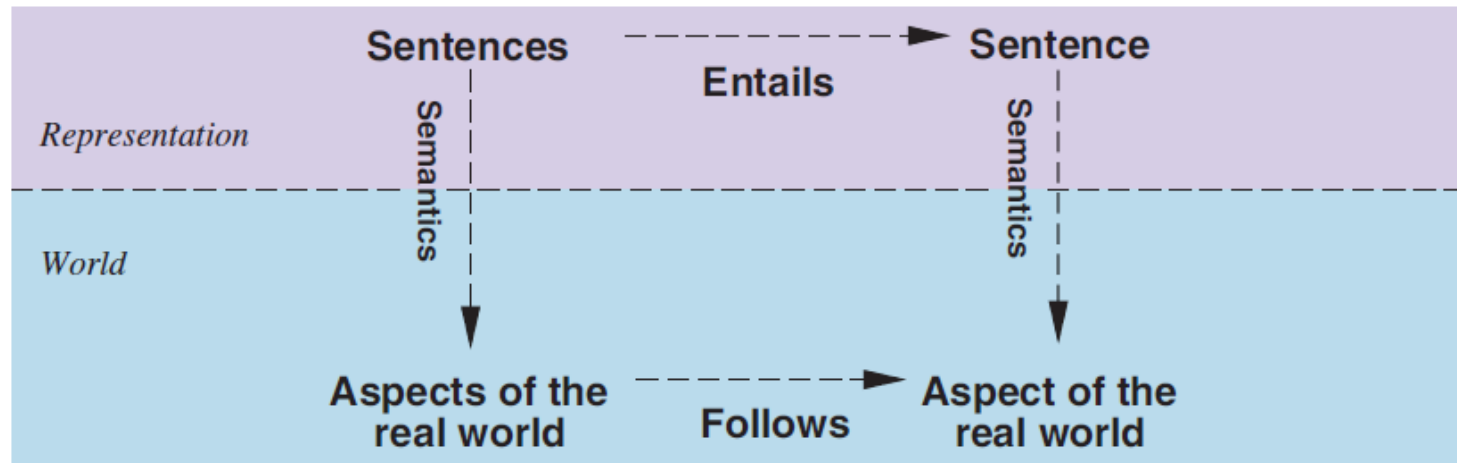
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- ♣ It is *complete* if it can derive any sentence that is entailed, that is,

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Logical Reasoning

A process whose conclusions are guaranteed to be true in any world in which the premises (the *KB* in this case) are true.



Correspondence between world and representation